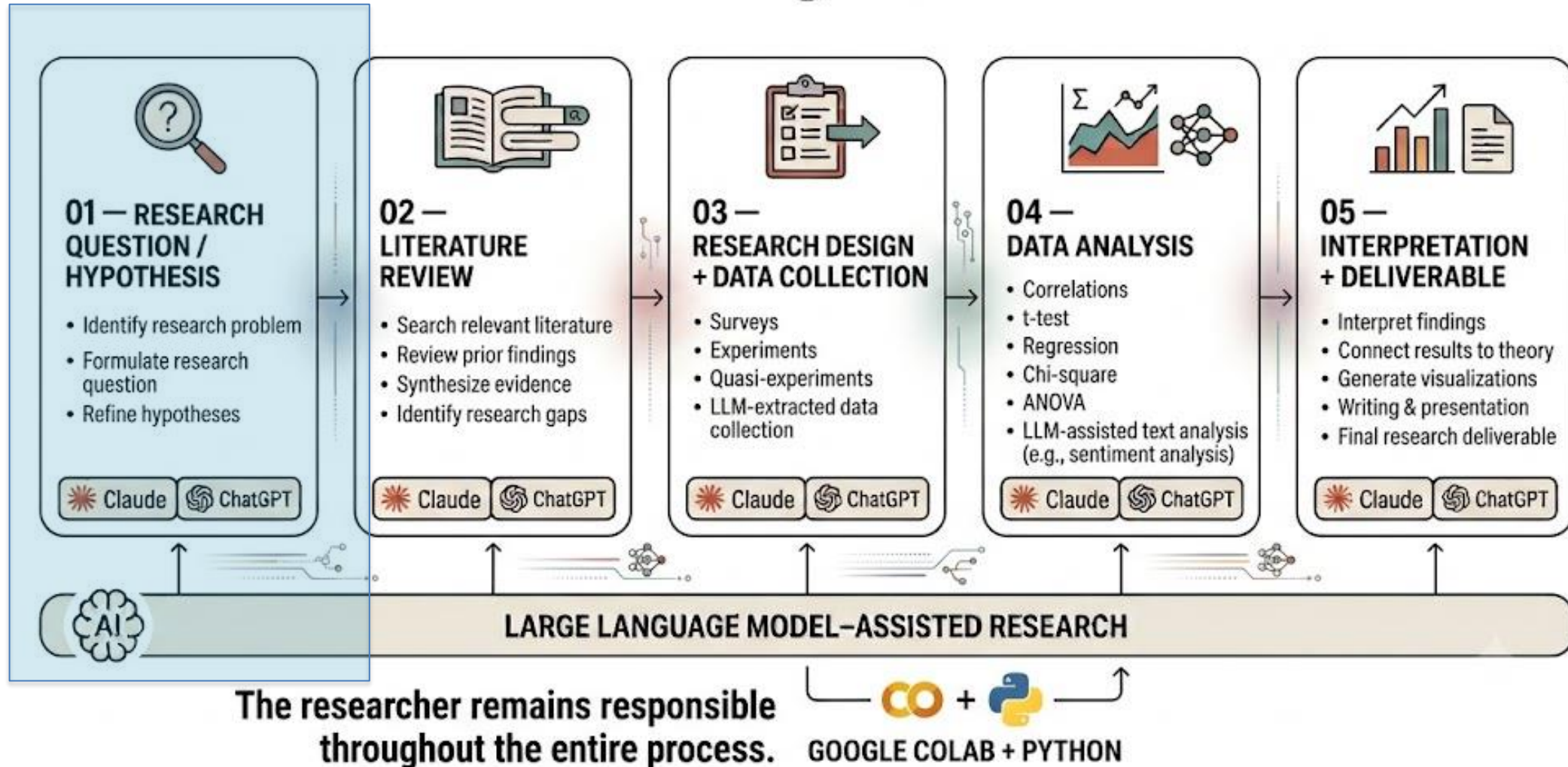


Fundamental Research Design: Research Questions, Hypotheses, Variables, and Surveys

BLOCK 2 • DAY 1 • STARTING FROM THE BASICS

Research Methods in the Age of AI



What is a research question?

- A research question states exactly what the study wants to find out.
- It should be specific enough that you can imagine collecting evidence for it.
- It usually tells us WHO or WHAT we care about.
- It often identifies something we want to describe, compare, or connect.

A good research question is answerable with data.



**A skincare brand paid for this post.
What might the brand want to KNOW?**

Think → pair → share
What would you want to know? What evidence
could answer it?

A topic is not yet a research question

Topic: Influencer marketing

RQ: Does seeing an #Ad disclosure change how credible an influencer seems?

Topic: Sports journalism

RQ: Do fans perceive athlete-written posts as more authentic than team-written posts?

Topic: Crisis PR

RQ: Is a CEO video apology associated with greater company trust than a written apology?

Topic: Advertising

RQ: How does humor in a social-media ad relate to purchase intention?

Quick check: topic or research question?

Call out TOPIC or RESEARCH QUESTION.

1. Instagram influencers
2. Does influencer expertise relate to followers' trust in a sponsored post?
3. Sports media
4. How often do college sports fans get game updates from social media?
5. Crisis communication

Three beginner-friendly kinds of research questions

DESCRIBE

How often do college students use TikTok for news?

RELATE

Is TikTok news use related to perceived news credibility?

COMPARE

Do people rate video apologies as more sincere than text apologies?

What is a hypothesis?

- A hypothesis is a **specific prediction** about **what you expect to find**.
- It usually predicts a relationship or a difference.
- It should be testable with the data you plan to collect.
- A hypothesis comes from prior research, theory, or a reasoned expectation—not just a random guess.

Research question = asks. Hypothesis = predicts.

Same study: research question vs. hypothesis

RQ

Do people rate influencer posts with #Ad as less authentic than posts without #Ad?

H1

People who see the #Ad version will report lower authenticity than people who see the no-#Ad version.

RQ

Is social-media sports use related to fan engagement?

H1

More frequent social-media sports use will be associated with higher fan engagement.

Research Question vs. Hypothesis

	Research Question	Hypothesis
Purpose	Asks what we want to know	Predicts what we expect to find
Form	Usually a question	Usually a statement
Direction?	May be open-ended	Often predicts direction/difference
Example	Does message tone relate to trust?	A warmer tone will be associated with greater trust.
Always needed?	Yes: every study needs a clear question	No: not every study needs a hypothesis

When should I use an RQ? When should I use a hypothesis?

- Use an RQ when you are asking what happens, how often, whether things are related, or whether groups differ.
- Use a hypothesis when you have a reasonable basis to predict the direction of the result.
- Descriptive studies often use RQs without hypotheses.
- Exploratory studies may use RQs because the expected pattern is not yet clear.

A study can have BOTH: an RQ states the question; a hypothesis states the predicted answer.

Do not invent a hypothesis just because you think every paper needs one.

RQ or hypothesis?

For each statement, decide: RQ or HYPOTHESIS?

1. Does an AI-generated label affect perceived ad credibility?
2. Ads labeled AI-generated will receive lower credibility ratings than unlabeled ads.
3. How often do students follow sports journalists on social media?
4. Fans who follow more sports journalists will report greater knowledge of team news.

What makes RQs and hypotheses researchable? VARIABLES

- Look at: “Is seeing an #Ad label related to how authentic an influencer seems?”
- Two things can differ:
 - #Ad disclosure: #Ad label present. #Ad label not present
 - Perceived authenticity: People may perceive the influencer as more or less authentic. We could measure this with an authenticity rating
- These changing characteristics are called **variables**.
 - Variable 1: #Ad disclosure
 - Variable 2: Perceived authenticity
- Variables are the building blocks that turn a research question into something we can measure and study.

If nothing can vary, there is nothing to measure or compare.

What is a variable?

- A variable is a characteristic that can take different values across people, posts, messages, organizations, or situations.
- For example, *Platform* can be Instagram, TikTok, YouTube, or X.
- *Number of likes* can be 0, 10, 500, or 50,000.
- *Trust* can be low, medium, or high—or measured on a rating scale.
- *#Ad disclosure* can be present or absent.

Variable = something that can vary.

Find the variables

RQ: “Do fans perceive player-created sports posts as more authentic than team-created posts?”

1. What is the first thing that varies?
2. What is the outcome that varies?
3. What values could each variable take?
4. Could we record those values in a dataset?

Types of Variables: categorical vs. numeric

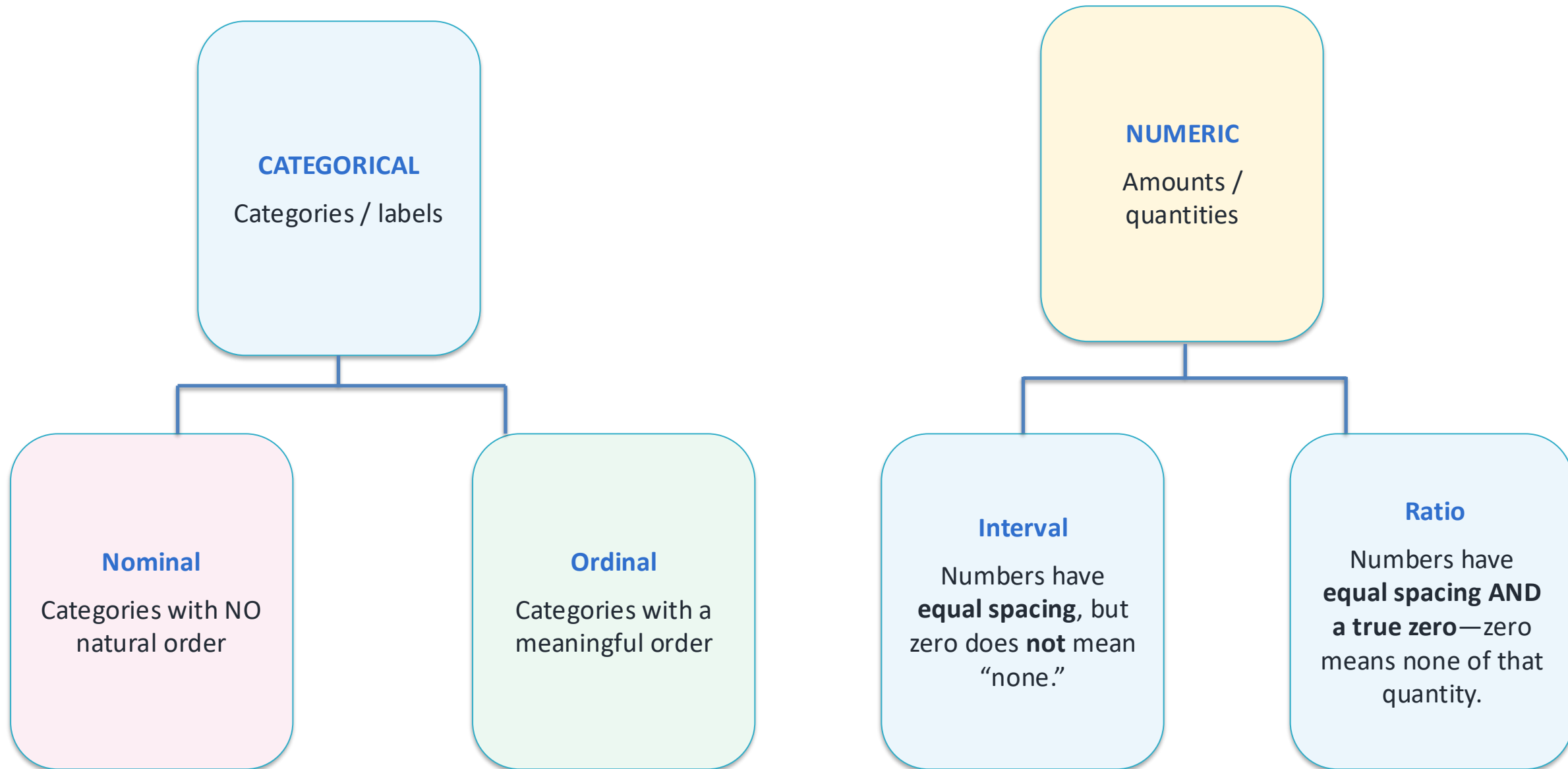
CATEGORICAL

Values tell us which group/category. Example:
platform = Instagram / TikTok / YouTube.

NUMERIC / QUANTITATIVE

Values represent an amount or quantity.
Example: **number of comments** = 0, 12, 245.

Types of Variables



1. Nominal variables: categories with no natural order

- The values are names or labels.
- There is no meaningful “higher” or “lower” category.
- Social platform: Instagram / TikTok / YouTube / X
- Content format: video / image / carousel
- Sports beat: football / basketball / baseball
- #Ad disclosure: yes / no

Nominal asks: WHICH category?

2. Ordinal variables: categories with an order

- The values can be ranked from lower to higher.
- But the distance between categories is not guaranteed to be equal.
- Brand reputation: Poor / Fair / Good / Excellent
- Frequency category: Never / Rarely / Sometimes / Often / Always
- Agreement item: Strongly disagree → Strongly agree

Ordinal asks: WHICH level or rank?

3. Interval variables: equal numerical intervals, but no true zero

- The difference between 10 and 20 represents the same amount as the difference between 20 and 30.
- But zero does NOT mean the complete absence of the thing.
- Classic example: temperature in °F or °C.
- Interval variables are less common in everyday communication examples than nominal, ordinal, or ratio variables.

Interval = equal spacing; zero is not “none.”

4. Ratio variables: equal intervals + a true zero

- Zero means none of the quantity is present.
- Number of Instagram comments: 0 really means no comments.
- Video viewing time: 0 seconds means no viewing time.
- Number of followers: 0 means no followers.
- Number of articles shared: 0 means none shared.

Ratio = equal spacing + meaningful zero.

Put the four levels together

Level	Broad family	Easy question	Communication example	Example values
Nominal	Categorical	Which category?	Primary social platform	Instagram / TikTok / YouTube
Ordinal	Categorical	Which ordered level?	Frequency of checking sports news	Never / Rarely / Sometimes / Often / Always
Interval	Numeric	How much, equal spacing, no true zero?	Temperature at outdoor PR event	50°F / 60°F / 70°F
Ratio	Numeric	How much, with a true zero?	Number of comments	0 / 5 / 200

What about 1–5 or 1–7 ratings?

- Example: “How credible is this post?” 1 = Not at all credible → 7 = Extremely credible.
- A single Likert-type item is technically ORDINAL: the response options have an order.
- The numbers are labels for ordered response categories; we should not automatically assume equal psychological distance between every point.
- Researchers often combine multiple related items into a scale and may analyze the resulting score as approximately continuous.

For this class: first identify what the response options actually mean.

A number printed beside a response option does not automatically make the variable ratio or interval.

Variable classification challenge

Classify each as nominal, ordinal, interval, or ratio.

1. Platform used most: Instagram / TikTok / YouTube
2. Agreement: Strongly disagree → Strongly agree
3. Number of likes on a post
4. Seconds spent watching a TikTok
5. Temperature during an outdoor football press conference

A variable name is not enough: make it measurable

Variable: “**engagement**” — what exactly does that mean?

- Could mean number of likes.
- Could mean number of comments or shares.
- Could mean click-through rate.
- Could mean seconds spent viewing.
- Could mean a self-report rating of attention or involvement.

Measurement = deciding exactly how an idea will be represented by observable data.

What is Measurement?

Measurement = deciding how we will turn a variable into data we can actually record.

VARIABLE

Trust in an influencer

"Trust" is an idea.
We cannot put the idea itself directly
into a dataset.



MEASURE IT

"I trust this influencer."

1 = Strongly disagree
↓
7 = Strongly agree



DATA

2 5 6
4 7 3

Now each participant has a
recorded value.

One concept can be measured in different ways

TRUST

Single 1–7 rating • multiple survey items • willingness to rely on source

ENGAGEMENT

Likes • comments • shares • viewing time • self-reported involvement

PURCHASE INTENTION

1–7 likelihood rating • click on “Shop Now” • choice between products

NEWS USE

Minutes per day • days per week • platform frequency categories

Common measurements in communication research

Category choice

Which platform do you use most? Instagram / TikTok / YouTube / Other

Yes / No

Did you notice the #Ad disclosure? Yes / No

Likert-type agreement

"I trust this influencer." Strongly disagree → Strongly agree

Rating scale

How credible is this story? 1 = Not at all → 7 = Extremely

Frequency

Never / Rarely / Sometimes / Often / Always

Behavioral / digital measure

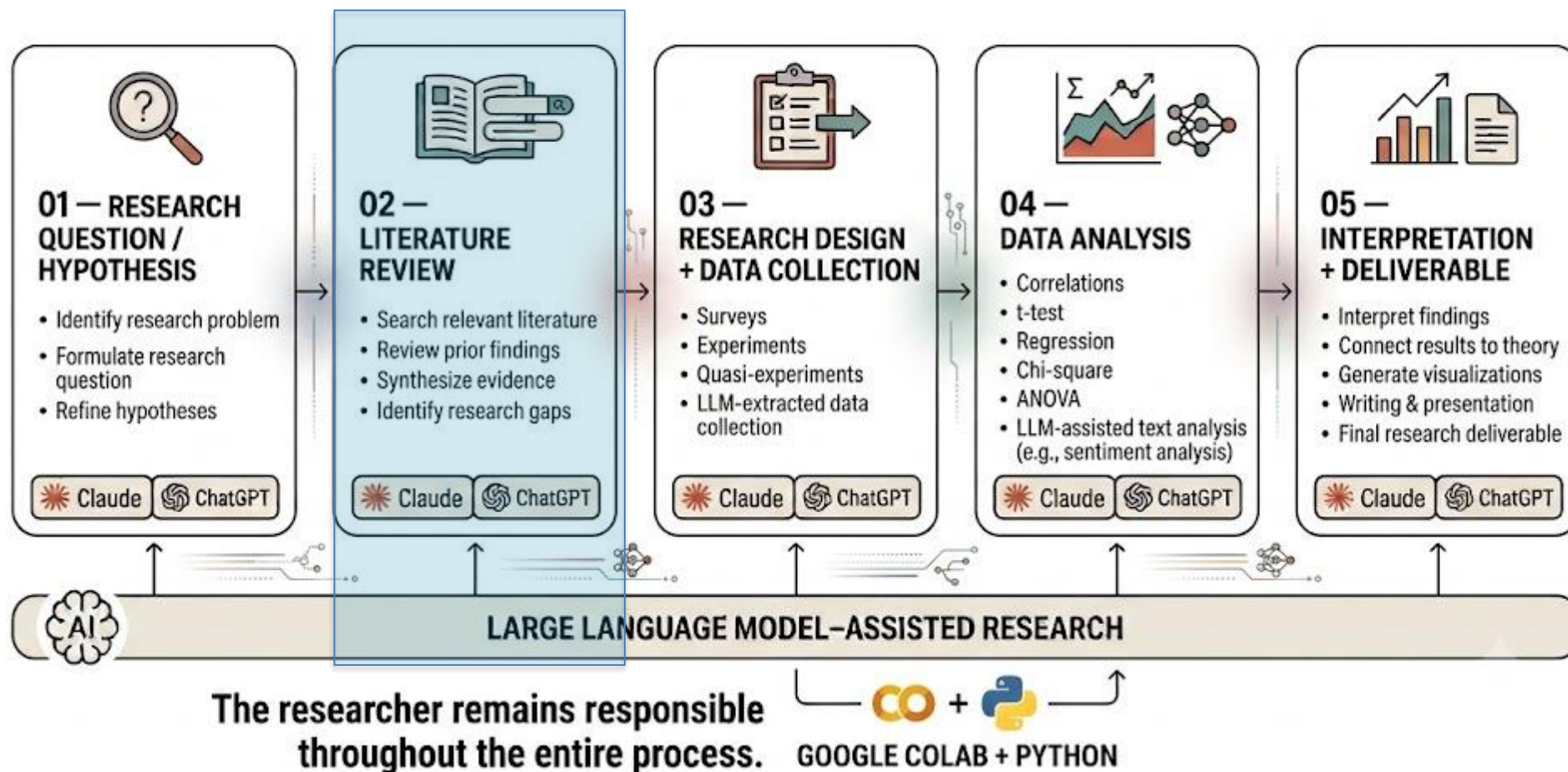
Clicks • shares • comments • viewing time

Measurement choices create variable types

Question / measure	Recorded values	Variable level
Which platform do you use most?	Instagram / TikTok / YouTube	Nominal
How often do you check team news?	Never → Always	Ordinal
How credible is this post?	1–7 ordered rating	Ordinal (single item)
How many comments did the post receive?	0, 1, 2, 3...	Ratio
How long did the person watch?	0.0, 3.4, 18.7 sec	Ratio

You have a research question and measurement. Now what?

Research Methods in the Age of AI



For example:

RQ: Does seeing an #Ad disclosure change how credible an influencer seems?

- Has someone already studied this?
- What did previous studies find?
- What do we still NOT know?
- Does our question add something new?

Literature review = finding, reading, and synthesizing previous research related to your research question.

What Does a Literature Review Actually Do?

1. Shows what we already know

What have researchers already found about #Ad disclosures, credibility, authenticity, or purchase intention?

2. Shows how others studied it

Who did they study? What questions or measures did they use? Instagram, TikTok, or another platform?

3. Finds the gap

Maybe prior work studied adults but not college students, Instagram but not TikTok, or found conflicting results.

4. Helps design our study

The literature helps refine the RQ, identify variables, develop hypotheses, choose measures, and select a design.

A literature review connects YOUR question to what researchers already know.

From Research Question → Literature → Study

1. RESEARCH QUESTION

Does #Ad disclosure affect influencer credibility?

2. SEARCH LITERATURE

influencer marketing • #Ad • disclosure • credibility

3. WHAT DO WE KNOW?

What has been found? What measures were used? What is missing?

4. REFINE STUDY

Variables + hypothesis

5. DESIGN

Survey? Experiment? Another design?

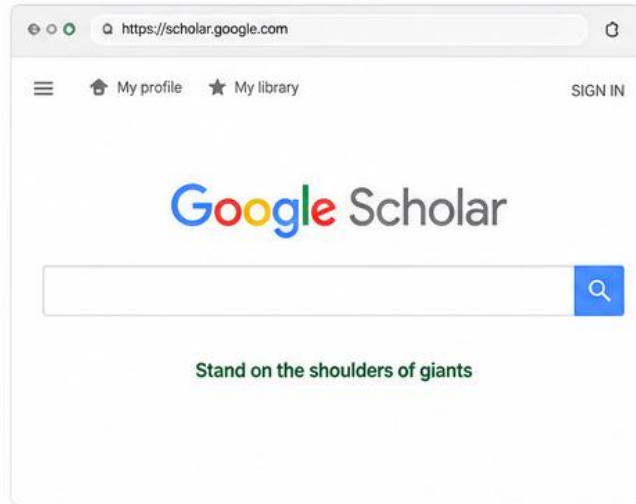
RQ → Literature Review → Variables & Hypothesis → Measurement → Research Design → Data

Finding Research with Google Scholar

A step-by-step example for your literature review

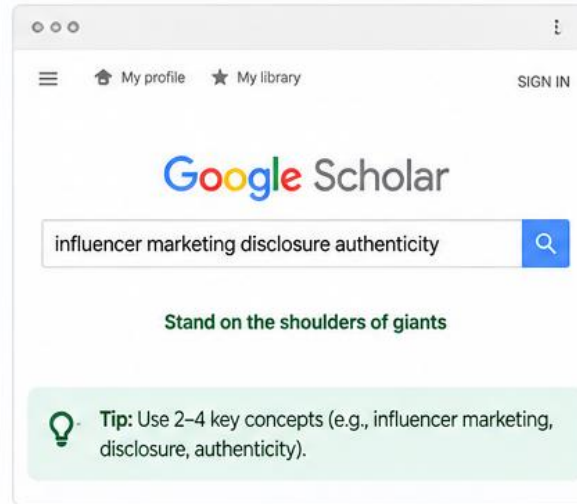
1 Go to Google Scholar

Visit <https://scholar.google.com>



2 Enter your search terms

Use keywords related to your topic.



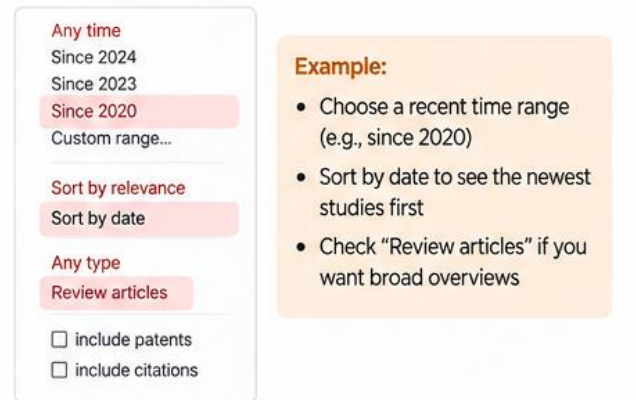
3 Review the results

Scan the titles and abstracts to find relevant studies.



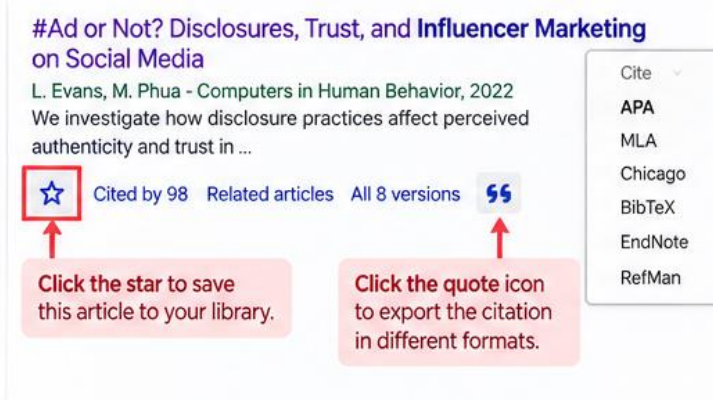
4 Use filters to narrow your search

You can filter by year, sort by date, or include only review articles.



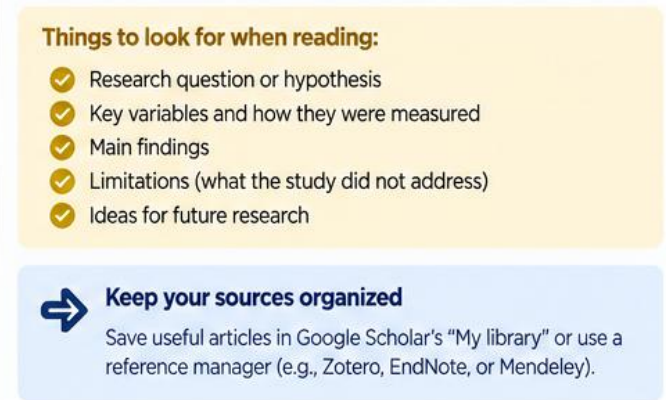
5 Save and organize useful articles

Click the star to save to "My library." You can also export citations.



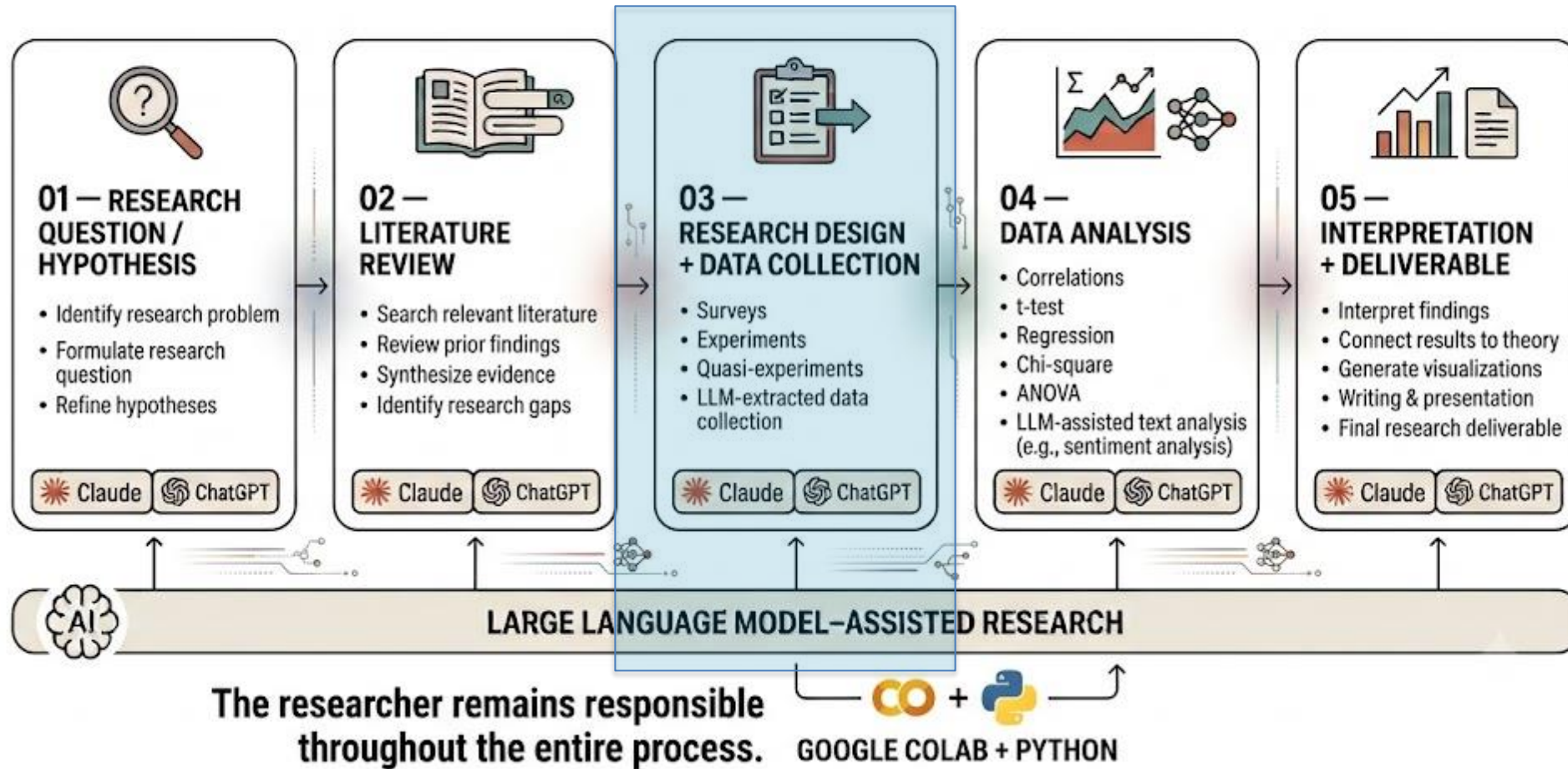
6 Read, take notes, and track your sources

Open the full text (if available), read the abstract, and keep track of key findings for your literature review.



Key Takeaway: Google Scholar is a free and powerful tool to find scholarly research. Use it to learn what is already known, identify gaps, and support your research question.

Research Methods in the Age of AI



Research design helps us to think about how to collect variables

What does “research design” mean?

- Research starts with a question. We decide what evidence would help answer the question.
- We decide what information must be collected.
- We decide who or what we will collect it from.
- Research design is **the PLAN connecting the question to the evidence.**

Question → evidence needed → plan to collect evidence (RESEAECH DESIGN) → answer

Survey is a research design; it's one way to collect variables

- A survey asks people a standardized set of questions.
- Surveys are especially useful for attitudes, beliefs, perceptions, intentions, self-reported behaviors, and demographics.
- A survey can include nominal, ordinal, and numeric variables in the same questionnaire.
- Survey is a RESEARCH DESIGN AND A DATA-COLLECTION METHOD; what you can conclude depends on the overall design.

Survey question → response → variable value → dataset

Example: A tiny social-media survey

Platform — nominal

Which platform do you use most? Instagram / TikTok / YouTube / X / Other

Frequency — ordinal

How often do you get news from social media? Never → Always

Credibility — ordinal item

This news post is credible. Strongly disagree → Strongly agree

Behavior — ratio

How many news posts did you share in the past 7 days? ____

Exposure — nominal

Have you seen this campaign before? Yes / No

Viewing time — ratio

How many minutes per day do you estimate using TikTok?

Good survey questions are boring—in a good way

Clear: Participants should interpret the question similarly.

Specific: define the time period when needed.

Neutral: do not push respondents toward an answer.

One idea at a time: avoid double-barreled questions.

Response options should match the question and cover reasonable answers.

Good measurement is more important than clever wording.

Bad question → better question

BAD: “How amazing was this ad?”

BETTER: “How favorable is your overall evaluation of this ad?” 1 = Very unfavorable → 7 = Very favorable

BAD: “How trustworthy and likable is this influencer?”

BETTER: Ask trust and likability as separate items.

BAD: “Do you use TikTok a lot?”

BETTER: “On a typical day, about how many minutes do you spend on TikTok?”

BAD: “Don’t you agree the CEO gave a sincere apology?”

BETTER: “The CEO’s apology seemed sincere.” Strongly disagree → Strongly agree

Activity: build a study from this post



1. Write ONE research question about this post.
2. Identify the variable(s) in your RQ.
3. Classify each variable: nominal / ordinal / interval / ratio.
4. Decide exactly how each variable will be measured.
5. If appropriate, write a hypothesis.
6. Write TWO survey questions that would collect the data.

The fundamental research-design chain

